

AKHIL BAJA

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PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years of experience building production-grade GenAI, ML and developer infrastructure at **Qualcomm** and **Axis Bank**. Engineered cross-language LLM context pipelines cutting retrieval latency by **60%** and improving LLM accuracy by **46%**, alongside enterprise GenAI systems serving **1,000+** users. Strong in Python, LLMs, RAG, machine learning, and distributed systems; MSc student in Artificial Intelligence Applications and Innovation at **Imperial College London**.

WORK EXPERIENCE

Qualcomm | Software Engineer (GenAI / Systems)

Dec 2024 – Aug 2026

Technologies: Python, C++, Jedi, Jenkins, RabbitMQ, Docker, Rancher, SQL, Git

- Engineered a cross-language **LLM context engineering** pipeline using AST-based dependency extraction across Java, Python, C and C++, reducing retrieval latency by **60%** and improving LLM accuracy by **46%** — **Qualcomm Certificate of Recognition**.
- Re-architected a central data-harvesting pipeline for **1,000+** users, cutting deployment failure rate by **84%** by replacing legacy Jenkins workflows with asynchronous **RabbitMQ** queues and containerizing microservices via **Docker/Rancher** for cross-OS deployment stability.
- Slashed static analysis parser execution latency by **93%** and consolidated **200+** unit tests into a Linux-compatible test harness, cutting ongoing maintenance overhead by **60%**.

Axis Bank | Data Scientist (AI / ML)

July 2023 – Dec 2024

Technologies: Python, Azure OpenAI, PySpark, Cloudera ML, SAS EG, SQL, XGBoost

- Built and deployed an enterprise GenAI assistant using **Azure OpenAI and Cloudera ML**, serving **1,000+** active employees and reducing internal query resolution time by **70%**.
- Engineered an **Isolation Forest** anomaly detection model for Gold Loans, flagging branch-level risk patterns tied to **£30M+** in projected annual portfolio loss exposure and cutting manual review time by **80%** — **BIU Star Award**.
- Automated customer Dynamic Risk Rating via **XGBoost and Naive Bayes** pipelines, reducing manual processing effort by **~40%** and aligning the automated risk-rating process with **RBI (Reserve Bank of India)** regulatory requirements.

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java

AI/ML & GenAI: Generative AI, Large Language Models (LLMs), RAG, LLM Context Engineering, Machine Learning, Deep Learning, Anomaly Detection, XGBoost, Naive Bayes, Computer Vision (U-Net, DenseNet)

Data & Platforms: PySpark, Azure OpenAI, Cloudera ML, PostgreSQL, SAS EG

Software & Infrastructure: Docker, Rancher, RabbitMQ, Jenkins, Git, AST Parsing, Static Analysis, Distributed Systems, Microservices, CI/CD, Unit Testing

EDUCATION

Imperial College London

Sept 2026 – Sept 2027

MSc in Artificial Intelligence Applications and Innovation

London, UK

IIT Hyderabad

Aug 2019 – May 2023

B.Tech in Engineering Science

CPI: 8.04 / 10.0

*Internship and Placement Cell Coordinator — Managed corporate outreach and recruitment operations for **100+** technology and engineering firms.*

PROJECTS

- Cosmic Ray Segmentation (Deep Learning):** Enhanced the **deepCR** model for cosmic-ray detection in astronomical images; evaluated **U-Net** and implemented residual connections, improving pixel-level segmentation precision by **15%**.
- Tuberculosis Detection Pipeline (Computer Vision):** Implemented a transfer-learning pipeline combining **U-Net** segmentation and **DenseNet** classification across **5,000+** chest X-rays (**+12%** accuracy over baseline).
- Scientific Publications Database (Data Infrastructure):** Designed and optimized a **PostgreSQL** database and custom ER schema for **2M+** research papers, accelerating query execution time by **45%** via targeted indexing.